Chapter 8: Mechanical Design of Overhead Lines

V.K. Mehta, Principles of Power System · Power System II (EEE 0713 3237)
Part 1: your derivations (digitised) · Part 2: every theory question with exam-ready answers

Part	What is inside	Source
1	D1 String efficiency (3-disc + 4-disc N.B.) · D2 Sag: definition, equal supports, unequal supports · D3 Effect of wind and ice loading · D4 Some formulas (working tension, clearance)	Your handwritten chap_8_derivations.pdf, digitised in the same steps
2	18 theory questions with simple answers, rated ★ to ★★★★★	Ch-8 Tier 1 + Tier 2 theory of the master suggestion, plus the teacher's Ch-8 lecture notes (Lec 11–12) and assigned Articles 8.1–8.15

Star rating: ★★★★★ asked 3 or more times in finals (learn first) · ★★★★ asked twice or flagged by the teacher · ★★★ asked once or in the lecture notes · ★★ in the assigned articles only.
Colour key: **Gold** = boxed in your notes / key formula · Purple = small addition by Claude · ⚠ = a slip to check.

Teacher's instruction (Lec-12): "CH-08 END here · Article 8.1, 8.2, 8.3, 8.4, 8.5, 8.6, 8.7, 8.8, 8.9, 8.10, 8.11, 8.12, 8.13, 8.14, 8.15". In class: "pin & suspension type ★★".

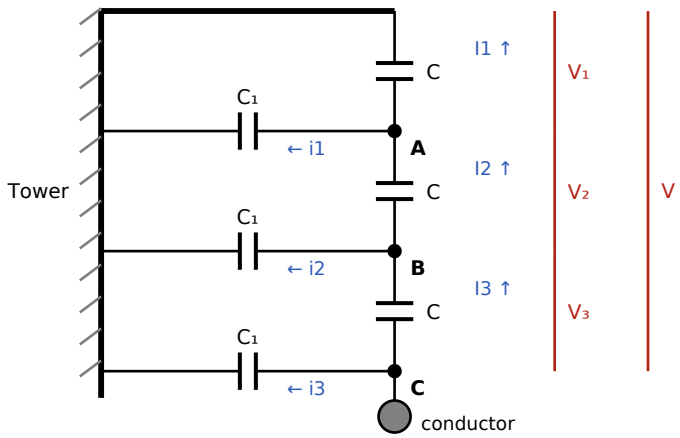
PART 1: DERIVATIONS (from your handwritten notes)

D1. Derivation: String Efficiency

Q: → derive string efficiency
→ potential distribution (V_1, V_2, V_3) + equivalent circuit
→ effect of K = 0.2, K = 0.8 and disc no.

C → self capacitance
C_1 → shunt capacitance
$K = C_1/C$

For 3-disc string:



Equivalent circuit of a 3-disc string

KCL at node A,

$I_2 = I_1 + i_1$ $\quad [I = V/X_C]$

$\Rightarrow \omega C \cdot V_2 = \omega C \cdot V_1 + \omega C_1 \cdot V_1$

$= \omega C \cdot V_1 + \omega \cdot (KC) V_1$

$= \omega C\, V_1 (1 + K)$

$\Rightarrow$ **$V_2 = V_1(1 + K)$**

and KCL at node B,

$I_3 = I_2 + i_2$

$\Rightarrow \omega C \cdot V_3 = \omega C\, V_2 + \omega C_1 (V_1 + V_2)$

$= \omega C\, V_2 + \omega (KC)(V_1 + V_2)$

$\Rightarrow V_3 = V_2 + K(V_1 + V_2)$

$= KV_1 + V_2(1 + K)$

$= KV_1 + V_1(1 + K)(1 + K)$

$= KV_1 + V_1(1 + K)^2$

$= V_1(K + 1 + 2K + K^2)$

$\therefore$ **$V_3 = V_1(1 + 3K + K^2)$**

Total voltage:

$V = V_1 + V_2 + V_3$

$= V_1 + V_1(1 + K) + V_1(1 + 3K + K^2)$

$\therefore$ **$V = V_1(3 + 4K + K^2)$**

String efficiency $= V/(3V_3) \times 100\ \%$

$= (3 + 4K + K^2) / [3(1 + 3K + K^2)] \times 100$

If K = 0, $\eta = (3 + 0)/(3(1 + 0)) \times 100 =$ **100 %**

So, voltage is shared equally.

K = 0.2, $\eta = 78\ \%$ $(V_2 = 1.2V_1, V_3 = 1.64V_1)$

So, K↑ → η↓ and disc nearest to conductor has maximum voltage.

($K = 0.8 \rightarrow V_2 = 1.8V_1, V_3 = 4.04V_1, \eta = 6.84/12.12 \times 100 \approx 56\ \%$: even lower, so K↑ → η↓.)

also, $\eta = V/(nV_n)$

$\therefore$ n↑ → η↓ (Shorter string has high η)

N.B.: If 4-string, at node C,

$\omega C \cdot V_4 = \omega C\, V_3 + \omega C_1 (V_1 + V_2 + V_3)$

$\Rightarrow V_4 = V_3 + K(V_1 + V_2 + V_3)$

$= V_1(1 + 3K + K^2) + K\, V_1(3 + 4K + K^2)$

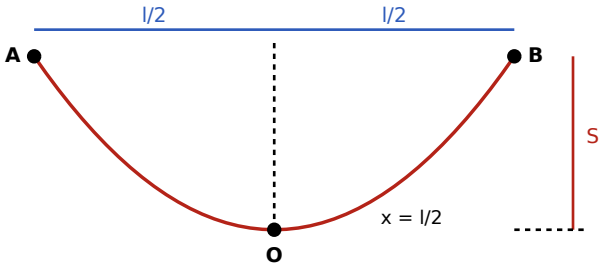
∴ **$V_4 = V_1(1 + 6K + 5K^2 + K^3)$**

∴ $\eta = V/(4V_4) \times 100$

D2. Sag in Overhead Line

Sag: The difference in level between point of supports and the lowest point on the conductor.

Sag: distance between support point and lowest point.



Sag calculation:

(i) When supports are at equal level:

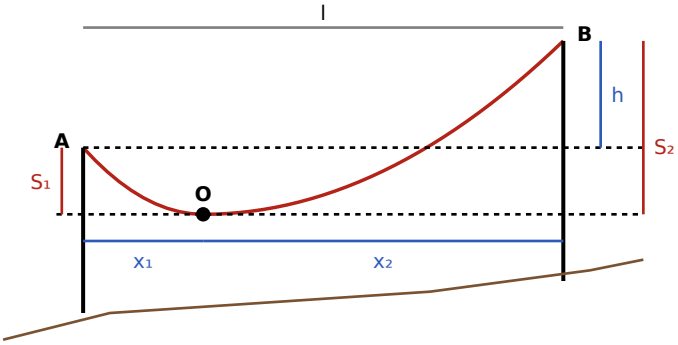
l = length of span

w = weight per unit length of conductor

T = tension in the conductor

∴ **$S = wl^2/8T$** [$S = wx^2/2T$, $x = l/2$]

(ii) When supports are at unequal level (in hilly areas):



l = span length

h = difference in level between supports = $S_2 - S_1$

x_1 = distance of lower level support from O

x_2 = " " higher " " O

T = tension of conductor

$x_1 + x_2 = l$ ———————— (1)

Sag, $S_1 = wx_1^2/2T$, $S_2 = wx_2^2/2T$

∴ $S_2 - S_1 = (w/2T)(x_2^2 - x_1^2)$

⇒ $h = (w/2T)(x_2 + x_1)(x_2 - x_1)$

⇒ $h = (w/2T)(l)(x_2 - x_1)$

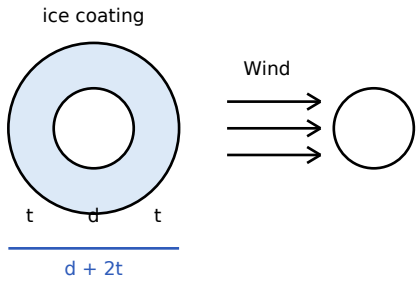
⇒ **$x_2 - x_1 = 2Th/wl$** ———————— (2)

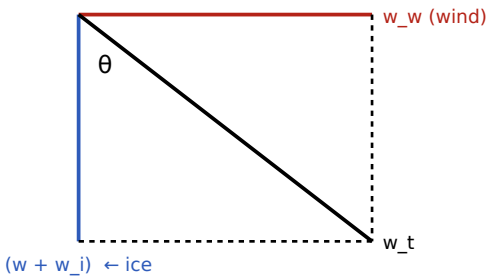
Solving eqn (1) and (2):

$x_1 = l/2 - Th/wl$

$x_2 = l/2 + Th/wl$

D3. Effect of Wind and Ice Loading





∴ **$w_t = \sqrt{((w + w_i)^2 + w_w^2)}$**

Where, w = density × volume per unit length (m = ρV)

w_i = density$_{ice}$ × πt(d + t) ← ice weight

w_w = **wind pressure × (d + 2t)** ← wind's weight / force

= wind pressure × projected area (= d × 1 (diameter) × 1, when there is no ice)

→ $\tan\theta = w_w / (w + w_i)$

→ slant sag, $S = w_t l^2/2T$; (in direction θ to the vertical)

⚠ for level supports use $S = w_t l^2/8T$ (= $w_t x^2/2T$ with x = l/2), as in the book Art 8.16 and your Type-7 maths.

→ vertical sag = S cosθ (S → slant sag)

D4. Some Formula

*** Working Tension, T = (Breaking stress × Conductor area) / Safety factor = Ultimate strength / Safety factor**

*** Tower related math:**

Minimum clearance: distance from the lowest point to the ground / water level.

Clearance midway: distance from the middle point to the water surface.

(Your Bengali notes for these two lines are given here in English.)

PART 2: THEORY QUESTIONS (exam-ready answers)

Write the answer in the same points. Draw the figure where one is given. "Asked" shows where the question came from.

Q1. What is string efficiency? Why is it not 100 %? Discuss the methods of improving string efficiency. ★★★★★

Asked: F21-4a, F22-5b, F24a-4b, older TT-2 (4 times) · Lec-11 Art 8.7 · **Book:** Mehta Art 8.7–8.8, p.168–171

Definition: The ratio of voltage across the whole string to the product of number of discs and the voltage across the disc nearest to the conductor is known as string efficiency.

String efficiency, $\eta = V / (n \times V_n) \times 100\ \%$, n = number of discs, V_n = voltage across the disc nearest to the conductor.

Why not 100 %: Shunt capacitance (C_1) exists between each metal link and the tower. So the charging current is not the same in every disc, and the disc nearest to the conductor gets the maximum voltage. Hence $\eta < 100\ \%$. (For d.c., capacitances are ineffective, so $\eta = 100\ \%$.)

Greater η → more uniform voltage distribution. 100 % is the ideal case (equal voltage on every disc).

Methods of improving string efficiency:

1. By using longer cross-arms: η depends on $K = C_1/C$. A longer cross-arm increases the distance of the string from the tower → shunt capacitance C_1 decreases → K decreases → η increases. Limit: cost and strength of the tower; in practice $K \approx 0.1$.

2. By grading the insulators (capacitance grading): Discs of different capacitance are used. The top unit (near the cross-arm) has the minimum capacitance, and capacitance increases towards the bottom unit (near the conductor). Since voltage $\propto$ 1/capacitance, the voltage becomes nearly equal. Drawback: many different sizes of discs are needed.

3. By using a guard ring (static shielding): A metal ring connected to the line conductor surrounds the bottom insulator. It creates capacitance between the metal fittings and the line conductor. The charging currents from the line cancel the pin-to-tower currents, so the same current flows through every disc → uniform voltage → η increases.

Q2. What is corona? Explain the theory of corona formation. ★★★★★

Asked: F21-5a, F22-3a, F24a-2a (3 times) · **Book:** Mehta Art 8.10, p.181–182

Definition: The phenomenon of violet glow, hissing noise and production of ozone gas in an overhead transmission line is known as corona.

It occurs when the applied voltage exceeds a certain value called the critical disruptive voltage. With d.c., the positive wire has a uniform glow and the negative wire has a spotty glow.

Theory of corona formation:

1. Air always contains some ionised particles (free electrons and +ve ions) due to cosmic rays, ultraviolet radiation and radioactivity.
2. When voltage is applied between the conductors, a potential gradient is set up in the air. It is maximum at the conductor surface.
3. Under this gradient the free electrons gain velocity. When the gradient reaches about **30 kV/cm (max)**, the electrons strike neutral molecules with enough force to knock out more electrons.
4. These new electrons and ions are also accelerated and ionise more molecules, so ionisation becomes cumulative.
5. The air around the conductor becomes conducting → corona is formed (or a spark takes place between the conductors).

Q3. Write down the advantages and disadvantages of corona. ★★★★★

Asked: F22-3a · **Book:** Mehta Art 8.13, p.184

Advantages:

1. Air around the conductor becomes conducting, so the virtual diameter of the conductor increases. This reduces the electrostatic stress between the conductors.
2. Corona reduces the effect of transients produced by surges (it acts like a safety valve).

Disadvantages:

1. Corona causes loss of energy → transmission efficiency decreases.
2. Ozone is produced, which may corrode the conductor by chemical action.
3. Corona current is non-sinusoidal → non-sinusoidal voltage drop → inductive interference with nearby communication lines.

Q4. Discuss the methods of reducing corona effect. ★★★★★

Asked: F21-5a · **Book:** Mehta Art 8.14, p.184

Corona is serious at 33 kV and above, so lines and bus-bars must be designed to avoid it.

1. By increasing conductor size: a bigger conductor raises the voltage at which corona occurs. This is one reason why ACSR conductors (larger cross-section) are used in transmission lines.

2. By increasing conductor spacing: a larger spacing raises the corona voltage. But spacing cannot be increased too much, because the cost of the supporting structure (bigger cross-arms, supports) increases.

Also helpful: smooth, clean conductors and bundled conductors (larger effective radius).

Q5. Define critical disruptive voltage. Write down its significance. ★★★★★

Asked: F22-4b, F24a-4a, F24b-1a (3 times) · **Book:** Mehta Art 8.12, p.182–183

Definition: Critical disruptive voltage is the minimum phase-neutral voltage at which corona occurs.

$V_c = m_0 g_0 \delta r \ln(d/r)$ kV/phase

g_0 = breakdown strength of air = 30 kV/cm (max) = 21.2 kV/cm (rms) · δ = air density factor = 3.92b/(273 + t) (= 1 at 76 cm Hg and 25 °C) · m_0 = irregularity factor (1 for polished, 0.98–0.92 for dirty, 0.87–0.8 for stranded conductors) · r = conductor radius, d = spacing.

Significance:

1. It tells the designer the voltage limit: the line must work below V_c to avoid corona.
2. Above V_c, corona power loss starts and increases as $(V - V_c)^2$, along with radio interference and ozone.
3. Conductor size (r) and spacing (d) are chosen so that the operating voltage is below V_c.
4. V_c falls in stormy weather and at low air density (hills), so a safety margin is kept.

Q6. What are the causes of insulator failure? (Explain flash-over and puncture.) ★★★★☆

Asked: F21-4c, F24a-5a (2 times) · **Book:** Mehta Art 8.5 (pin type), p.166

Insulators must withstand both mechanical and electrical stresses. Electrical breakdown happens in two ways:

1. Flash-over: An arc occurs between the line conductor and the insulator pin (earth) through the air gap around the insulator. The insulator can still work after flash-over unless the heat of the arc destroys it.

2. Puncture: The discharge occurs from the conductor to the pin through the body of the insulator. The insulator is permanently destroyed by the heat.

To avoid puncture, sufficient thickness of porcelain is provided. **Safety factor = Puncture strength / Flash-over voltage** is kept high (about 10 for pin type), so that flash-over happens before puncture.

Other causes: cracks, porosity or impurities in the porcelain; dirt and salt deposits on the surface; excessive mechanical load; sudden temperature changes; lightning surges.

Q7. How does electrical breakdown happen in a pin-type insulator? ★★★★☆

Asked: F24b-3a · **Book:** Mehta Art 8.5, Fig. 8.6, p.166

Breakdown of a pin-type insulator occurs by **flash-over** or **puncture**.

Flash-over: the arc jumps across the air gaps between the conductor and the pin, following the shortest arcing distance (a + b + c in Fig. 8.6). The insulator usually survives.

Puncture: the discharge passes through the body of the porcelain from the conductor to the pin; the insulator is permanently damaged.

So the porcelain is made thick enough that the puncture strength is about 10 times the flash-over voltage (safety factor ≈ 10).

Q8. What are the different types of insulators? Why are suspension insulators preferred above 33 kV? ★★★★☆

Asked: Lec-11 Art 8.5 (teacher: pin & suspension ★★); book review Q5, discussion Q5 · **Book:** Mehta Art 8.5, p.165–167

1. Pin type (< 33 kV): fixed on the cross-arm by a galvanised steel pin; the conductor sits in a groove on top and is tied with annealed wire. Above 33 kV it becomes too bulky and costly.

2. Suspension type (> 33 kV): a string of porcelain discs joined by metal links; the conductor hangs at the bottom and the top is fixed to the cross-arm. Each disc is designed for about 11 kV (e.g. 66 kV → 6 discs).

3. Strain insulator: used at a dead end, corner or sharp curve where the line has high tension. For low voltage (< 11 kV) shackle insulators are used; for high voltage, suspension discs are assembled in the vertical plane (two or more strings in parallel for long river spans).
4. Shackle insulator: used for low-voltage distribution lines; fixed to the pole or cross-arm with a bolt, in a horizontal or vertical position.
Why suspension insulators above 33 kV (advantages):
1. Cheaper than pin type beyond 33 kV.
2. Each disc is for about 11 kV; the required number of discs can be added for any voltage.
3. A damaged disc can be replaced; the whole string is not wasted.
4. Gives flexibility: the string can swing and take the position of minimum mechanical stress.
5. Line voltage can be raised later just by adding discs.
6. Conductors hang below the earthed cross-arm, which gives partial protection from lightning.

Q9. Explain the potential distribution over a suspension insulator string. Why is it not uniform? ★★★★☆

Asked: Lec-11 Art 8.6; book review Q8, Q10 · **Book:** Mehta Art 8.6, p.168

Each porcelain disc lies between two metal links, so each disc forms a capacitor C, called the **mutual or self capacitance**.
Capacitance also exists between the metal fitting of each disc and the tower (earth). This is the **shunt capacitance C_1**. (Teacher's note: pole and conductor behave like conductors, while air and the insulation disc behave like insulators, so capacitance forms between them.)
If only self capacitance existed, the same charging current would flow through all discs and each disc would get V/3. Because of shunt capacitance, the charging current is **not** the same in all discs, so the voltage across each disc is different.
Points to note: (i) voltage is not uniformly distributed due to shunt capacitance; (ii) the disc nearest to the conductor has the maximum voltage, and voltage decreases towards the cross-arm; (iii) the unit nearest the conductor is under maximum stress and is likely to be punctured; (iv) for d.c., voltage is equal across every unit.
(Draw the 3-disc equivalent circuit from D1.)

Q10. Define sag. Why is sag necessary? What is a stringing chart? ★★★★☆

Asked: F22-4a, F24b-4a (stringing chart) · **Book:** Mehta Art 8.15, p.186–187; Art 8.17(iv), p.199

Sag: The difference in level between the points of supports and the lowest point on the conductor is called sag.
Why sag is provided: If the conductor is stretched fully, the tension becomes too high and it may break. So conductors are allowed a dip (sag) to keep the tension within safe limits.
Low sag → tight wire → high tension (risk of breaking). High sag → low tension but more conductor material, taller supports and less ground clearance. So a compromise is made.
Sag depends on the conductor weight, span length, tension, temperature, and wind and ice loading.
Stringing chart: curves of sag vs temperature and tension vs temperature, plotted for a given conductor and loading. They are used in the field while stringing the conductors (as temperature rises, sag increases and tension decreases).

Q11. What are the main components of an overhead line? ★★★★☆

Asked: F22-5a · Lec-11 · **Book:** Mehta Art 8.1, p.160

1. **Conductors:** carry electric power from the sending end to the receiving end.
2. **Supports:** poles or towers that keep the conductors at a suitable height above the ground.
3. **Insulators:** attached to the supports; insulate the conductors from the ground.
4. **Cross arms:** provide support to the insulators.
5. **Miscellaneous items:** phase plates, danger plates, lightning arrestors, anti-climbing wires, earth wire.

(Teacher's figure: tower with cross arm, insulators, conductors and an earth wire on top.)

Q12. What are the desirable properties of insulators? ★★★☆☆

Asked: Older TT-2 · book review Q4 · **Book:** Mehta Art 8.4, p.164–165

Insulators keep the line current from flowing to earth through the supports. They should have:

1. High mechanical strength to withstand conductor load, wind load etc.
2. High electrical resistance to avoid leakage current to earth.
3. High relative permittivity so that dielectric strength is high.
4. Material should be non-porous and free from impurities and cracks.
5. High ratio of puncture strength to flash-over.

The most commonly used material is **porcelain** (also glass and steatite). Porcelain is mechanically stronger than glass, gives less trouble from leakage and is less affected by temperature changes.

Q13. Properties of conductor materials. Between aluminium and copper, which one is superior and why? (Why ACSR?) ★★★☆☆

Asked: F21-1b, older TT-1 · Lec-11 · book discussion Q6 · **Book:** Mehta Art 8.2, p.160–162

Properties of a good conductor material: (i) high electrical conductivity; (ii) high tensile strength; (iii) low cost; (iv) low specific gravity (light weight). Examples: copper, aluminium, steel-cored aluminium (ACSR), galvanised steel, cadmium copper.

Copper: high conductivity and tensile strength; high current density, so a smaller cross-section is needed and the wind load is less; durable, high scrap value. But it is costly and less available.

Aluminium: cheap and light (2.71 g/cc vs 8.9 g/cc → about half the weight of copper), so supports can be lighter. But its conductivity is 60 % of copper's, so the diameter must be about 1.26 times larger (more wind load); its tensile strength is lower and its expansion higher, so sag is greater and larger cross-arms are needed.

Conclusion: considering cost, conductivity, strength and weight together, **aluminium is superior (economical)** and is widely used today, especially as ACSR.

ACSR (Aluminium Conductor Steel Reinforced): a central core of galvanised steel wires surrounded by aluminium strands. Steel gives the tensile strength and aluminium carries the current. Advantages: less sag, so longer spans and shorter towers can be used.

Q14. What are the properties of line supports? Discuss the different types of line supports. (Which supports are used for distribution up to 11 kV?) ★★★☆☆

Asked: F24b-3b · Lec-11 Art 8.3 (teacher's table) · book review Q3 · **Book:** Mehta Art 8.3, p.162–164

Properties: 1. high mechanical strength; 2. light in weight; 3. cheap in cost and economical to maintain; 4. longer life; 5. easy access to conductors for maintenance.

1. Wooden poles: made of seasoned wood; span about 50 m; cheap and easily available; used for distribution in rural areas. Problems: tend to rot below ground level; life only 20–25 years; cannot be used above 20 kV; less mechanical strength; need periodic inspection.

2. Steel poles: substitute for wooden poles; greater mechanical strength, longer life, longer spans; used for distribution in cities; need galvanising or painting.

3. RCC poles (reinforced concrete): greater mechanical strength, longer life and longer spans than steel poles; little maintenance, good appearance. Problem: heavy, so transport cost is high (often made on site).

4. Steel towers: used for long-distance transmission at high voltage; greatest strength, longer spans, withstand severe weather; tower footings are earthed, so they protect against lightning; double-circuit towers keep the supply running if one circuit fails.

Distribution up to 11 kV: wooden, steel and RCC poles are used (steel towers are used for higher voltages).

Q15. What are the factors affecting corona? ★★★☆☆

Asked: Assigned Art 8.11 · book review Q12 · **Book:** Mehta Art 8.11, p.182

1. **Atmosphere:** in stormy weather the number of ions is higher, so corona occurs at a much lower voltage than in fair weather.
2. **Conductor size:** a rough, irregular (stranded) surface gives more corona than a smooth solid conductor; a larger diameter gives less corona.
3. **Spacing between conductors:** if spacing is very large compared to the diameter, there may be no corona.

4. **Line voltage:** corona forms only when the line voltage is high enough to make the air around the conductor conducting.

Q16. Explain visual critical voltage and power loss due to corona. ★★★☆☆

Asked: Assigned Art 8.12 · book review Q14 · **Book:** Mehta Art 8.12, p.183

Visual critical voltage (V_v): the minimum phase-neutral voltage at which the corona glow appears all along the line conductors. It is higher than V_c.

$V_v = m_v g_0 \delta r (1 + 0.3/\sqrt{(\delta r)}) \ln(d/r)$ kV/phase ; m_v = 1 (polished), 0.72–0.82 (rough).

Power loss due to corona (Peek's formula):

$P = (242.2/\delta)(f + 25) \sqrt{(r/d)} (V - V_c)^2 \times 10^{-5}$ kW/km/phase ; f = frequency, V = phase voltage, V_c = disruptive voltage.

Q17. Why are overhead lines preferred to underground cables for power transmission? ★★★☆☆

Asked: Lec-11 (intro of Ch-8) · Mehta introduction · **Book:** Mehta Ch-8 introduction, p.159

1. **Installation cost:** power is sent over long distances, and underground cables are much more expensive to install.
2. **High voltage:** it is very difficult to give cables enough insulation for very high voltages (insulation failure). In overhead lines bare conductors are used and air acts as the insulation.

So long-distance transmission is done by overhead lines; cables are used in congested cities where safety and appearance matter.

Q18. Important points while solving string-efficiency problems. ★★☆☆☆

Asked: Assigned Art 8.9 · **Book:** Mehta Art 8.9, p.171

1. The maximum voltage appears across the disc nearest to the conductor (line conductor).
2. Voltage across the string = phase voltage, i.e. voltage between line and earth.
3. **Line voltage = √3 × voltage across string**

Quick checklist (tick when learnt)

#	Question	Rating
1	What is string efficiency? Why is it not 100 %? Discuss the methods of improving string efficiency.	★★★★★
2	What is corona? Explain the theory of corona formation.	★★★★★
3	Write down the advantages and disadvantages of corona.	★★★★★
4	Discuss the methods of reducing corona effect.	★★★★★
5	Define critical disruptive voltage. Write down its significance.	★★★★★
6	What are the causes of insulator failure? (Explain flash-over and puncture.)	★★★★
7	How does electrical breakdown happen in a pin-type insulator?	★★★★
8	What are the different types of insulators? Why are suspension insulators preferred above 33 kV?	★★★★
9	Explain the potential distribution over a suspension insulator string. Why is it not uniform?	★★★★
10	Define sag. Why is sag necessary? What is a stringing chart?	★★★★
11	What are the main components of an overhead line?	★★★★
12	What are the desirable properties of insulators?	★★★
13	Properties of conductor materials. Between aluminium and copper, which one is superior and why? (Why ACSR?)	★★★
14	What are the properties of line supports? Discuss the different types of line supports. (Which supports are used for distribution up to 11 kV?)	★★★
15	What are the factors affecting corona?	★★★
16	Explain visual critical voltage and power loss due to corona.	★★★
17	Why are overhead lines preferred to underground cables for power transmission?	★★★
18	Important points while solving string-efficiency problems.	★★